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Mobile jammer enabled secure UAV communication with short packet transmission

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ABSTRACT

This paper studies a mobile jammer enabled secure unmanned aerial vehicle (UAV) communication network in the finite packet length scenario, where the external friendly jammer UAV will transmit jamming signal to deteriorate the quality of eavesdropping channel, and help to improve the secure communication performance of the source UAV. Specifically, an average effective secrecy rate (AESR) maximization problem is formulated by jointly designing the trajectories and transmit power of the source/jammer UAV subject to the UAVs' mobility constraints and power constraints. It is also verified that there exists a unique optimal packet error rate to maximize the AESR. The formulated problem is mathematically intractable, as it is nonconvex and involves coupled optimization variables. Therefore, we first decompose the trajectories and power optimization problem into four non-convex subproblems, and then transform them into standard convex problems by applying first-order Taylor expansion. Then, an efficient alternating optimization algorithm for AESR problem based on Bisection method and successive convex approximation (SCA) technology is presented by addressing the five subproblems iteratively. Simulation results are provided to verify that our proposed scheme can obtain considerable better AESR gain compared to other four benchmark schemes.

1. Introduction

Unmanned aerial vehicle (UAV) has attracted extensive attention in both military and civilian applications for its flexible on-demand deployment, fully-controllable mobility, low cost and stable line of sight (LoS) communication link [1–3]. Recently, UAV-enabled communication networks are of great significance in the field of wireless communication, which can be expected to play a considerable role in the six-generation (6G) communication systems [4,5]. Simultaneously, extensive research works have been concentrated on utilizing UAVs as various types of wireless communication platforms [6], such as mobile relays [7–9], mobile aerial base stations (BSs) and so on [10,11]. Especially, using UAVs as aerial mobile BSs can be regarded as a prospective solution to improve the communication performance of the cellular mobile communication systems.

Noting that wireless communications are susceptible to be intercepted by the eavesdroppers (Eve) for the characteristics of broadcast [12], which can bring about a new security problem. Consequently, physical layer security (PLS) has become as a desirable candidate to address the security issues in wireless communication [13,14]. A pivotal metric that has been extensively employed in PLS is the secrecy rate [15], which can reliably transmit confidential information (CI) and prevent eavesdroppers receive any CI about the message. Particularly, it can obtain a nonzero secrecy rate when the intensity of the eavesdropping communication channel is weaker than that of the legitimate communication channel. Generally speaking, the research relevant to beyond fifth-generation (B5G) wireless systems and PLS all supposed that CI utilizes infinite packet length. However, this hypothesis cannot be realized in practical application. Especially in the UAV-enabled communication networks, using infinite packet length will influence the reliability of the system performance because it is impracticable to transmit infinite packet length with an extraordinary small flight time slot [16,17]. Moreover, the finite packet length is provided with the characteristics of ultra-reliable and low-latency communication (URLLC), which can satisfy the requirement of B5G wireless communication [18].

Although the UAV network has many advantages, the data communication may suffer from transmission latency. Furthermore, the information security of legitimate ground receivers (GRs) will not be ensured when eavesdropping exists. To address these problems, short packet communication (SPC) can be utilized in the UAV communication

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system to obtain a lower transmission latency, and mobile jammer can be employed to enhance the security of UAV communication. Trajectory planning or power control has been a main research field in the previous literature on UAV-enabled networks [19–22]. In particular, Wu et al. [19] studied the UAVs' characteristics by optimizing user scheduling and association, trajectories, and power for a multi-UAV communication system; Zhang et al. [20] minimized the outage probability by designing the mobile device, the UAV's trajectory and power in the UAV relay communication network; the max–min average rate can be obtained by jointly designing the user scheduling and trajectory of UAV in Ref. [21], where the UAV is a mobile aerial BS to communicate with the legitimate users, and Pang et al. [22] maximized the average achievable rate of UAV relaying network by jointly optimizing the UAV trajectory, the IRS passive beamforming and transmit beamforming. In addition, in order to consider the importance of security performance, Dong et al. [23] proposed an alternating optimization algorithm to optimize the beamforming and resource allocation, which can obtain the maximal secrecy rate in the existence of an eavesdropper; Wang et al. [24] studied the throughput in UAV-aided NOMA communication by optimizing the power splitting radio, the time radio and the precoding vectors. In Ref. [25], Zhang et al. utilized PLS technique to maximize the transmission secrecy rate by jointly designing the legitimate user's transmit power and the UAV's trajectory. By utilizing the successive convex approximation (SCA) technique, Zhou et al. [26] maximized the minimum average secrecy rate by jointly designing the UAVs' trajectories and power. Different from conventional wireless network, the packet with finite packet length can effectively solve the problem of information transmission delay. Refs. [27,28] considered the communication performance based on the packet with finite packet length, Pan et al. [27] analyzed the packet error rate of the UAV-enabled relay system by jointly optimizing the packet length and UAV's location, and Wang et al. [28] studied the average effective secrecy rate (AESR) under the impact on the transmit power and trajectory of UAV.

As aforementioned, most of the literatures only research on the security performance of UAV-enabled communication without considering the impact of packet length, which is inapplicable for wireless communication with the low latency demand. Motivated by the above studies, we investigate the mobile jammer enabled secure UAV communication network with short packet transmission in this paper, where the source UAV (UAV S) transmits CI signal to legitimate GRs, and an external friendly jammer UAV (UAV J) will transmit jamming signal to degrade the quality of eavesdropping channel, and the security communication performance of legitimate channel can be enhanced accordingly. Considering the communication security of our proposed system, the AESR with finite packet length will be maximized by jointly optimizing the trajectories and transmit power of the UAV S/J under the conditions of the UAVs' mobility constraints and peak/average power constraints, in which we consider the influence of packet error rate. In order to achieve the optimal design, an alternating optimization algorithm for AESR problem based on Bisection method and successive convex approximate (SCA) technique is proposed in this paper. Furthermore, our proposed scheme can extremely enhance the AESR of the mobile jammer enabled secure UAV communication network compared with other four benchmark schemes. The main contributions of this work can be highlighted as follows.

• In UAV-enabled secure network, we study the AESR optimization problem in the presence of an eavesdropper within finite packet length, which is extraordinary different from the aforementioned papers with infinite packet length. On this basis, the influence of the UAV S/J's trajectory, transmit power and packet error rate on secure communication performance are considered comprehensively.

• To solve the non-convex and complicated AESR maximization problem, we first decompose the trajectory and power optimization problem into four nonconvex subproblems, and these four subproblems can be transformed into standard convex optimization problems by utilizing first-order Taylor expansion. Then, the concavity of the packet error rate can be studied comprehensively. On this basis, the AESR maximization problem can be obtained by jointly optimizing the trajectories and transmit power of the UAV S/J under the optimal packet error rate. Finally, an alternating optimization algorithm for AESR problem based on Bisection method and SCA technique is formulated to address the proposed optimization problem effectively.

• Simulation results demonstrate the superiority and effectiveness of our proposed alternating algorithm in improving the AESR of the mobile jammer enabled secure UAV communication network. The flying periods of the UAVs have an extremely impact on the system performance. Besides, it is shown that the AESR of our proposed scheme outperforms the other benchmark schemes obviously.

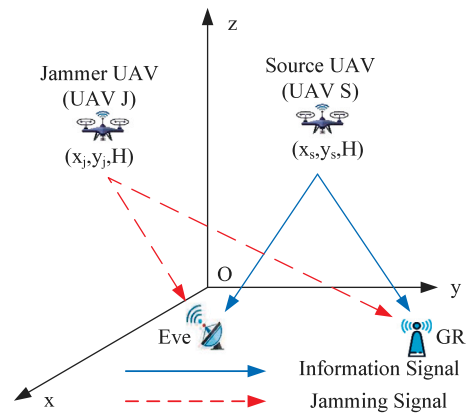


Fig. 1. Mobile jammer enabled secure UAV communication system.

The remainder of the paper is organized as follows. Section 2 presents the mobile jammer enabled secure UAV communication network system model and formulates the AESR maximization problem. An alternating optimization algorithm for AESR problem based on Bisection method and SCA technique is proposed to address the formulated optimization problem in Section 3. Compared with other four benchmark schemes, Section 4 provides simulation results to verify the superiority and effectiveness of our proposed algorithm. Finally, the conclusions are given in Section 5.

2. System model and problem formulation

As shown in Fig. 1, we investigate a mobile jammer enabled secure UAV communication system with finite packet length where the UAV S is utilized as an aerial base station to communicate with GR, in order to avoid CI signal being monitored by an eavesdropper, an external friendly UAV J will transmit interference signals to confuse Eve and protect CI transmission. Generally speaking, for better communication quality and lower information leakage, the flying period should be as long as possible, and the UAV S /J should fly to GR/Eve as close as possible. We assume that the location of Eve is known since it can be a legitimate GR in other multifarious substantial scenarios.

In secure UAV communication network, we consider a three-dimensional (3D) Cartesian coordinate system with GR and Eve located at $\mathbf{w}_g = [x_g, y_g, 0]^T$ and $\mathbf{w}_e = [x_e, y_e, 0]^T$ in meter (m), respectively. Define the trajectories of UAV S and UAV J as $\mathbf{q}_s(t) = [x_s(t), y_s(t), H]^T$ and $\mathbf{q}_j(t) = [x_j(t), y_j(t), H]^T$ with flight period T, where $0 \le t \le T$, and H is the minimum flight altitude for UAV to avoid buildings or obstacles. Furthermore, the Doppler effect caused by the mobility of UAVs can be well compensated. For the sake of simplicity, the flight period of UAV S/J can be divided into N time slots with equal duration, i.e., $\delta_t = T/N$, where δ_t is sufficiently small to ensure that the position of UAV can be regarded as unchanged in each time slot [19]. Accordingly, the UAVs' trajectories can be discretized into N-length sequences, i.e., $\mathbf{q}_s[n] = [x_s(n), y_s(n), H]^T$ and $\mathbf{q}_j[n] = [x_j(n), y_j(n), H]^T$,

$1 \le n \le N$. On this basis, the UAVs' mobility constraints (1) can be equivalently expressed as

$$\begin{aligned} &\left\|\mathbf{q}_s[n+1]-\mathbf{q}_s[n]\right\| \le V_{\max}\delta_t, \forall n \\ &\left\|\mathbf{q}_j[n+1]-\mathbf{q}_j[n]\right\| \le V_{\max}\delta_t, \forall n \\ &\mathbf{q}_s[1]=\mathbf{q}_{sI}, \mathbf{q}_s[N]=\mathbf{q}_{sF} \\ &\mathbf{q}_j[1]=\mathbf{q}_{jI}, \mathbf{q}_j[N]=\mathbf{q}_{jF} \end{aligned} \tag{1}$$

where $V_{\max}$ is the UAV S/J's maximum flight speed in m/s, $\mathbf{q}_s[1]/\mathbf{q}_j[1]$ and $\mathbf{q}_s[N]/\mathbf{q}_j[N]$ represent the initial and final positions of UAV S/J, respectively.

In time slot n, the Euclidean distance between UAV S/J and GR/Eve is given by

$$d_{xy}[n]=\left\|\mathbf{q}_x[n]-\mathbf{w}_y\right\|, x\in\{s,j\}, y\in\{g,e\}, \forall n \tag{2}$$

For simplicity, it is assumed that the communication links from UAV S/J to GR/Eve are dominated by line-of-sight (LoS) channel, and the air-to-ground (A2G) channel quality is mainly related to the distance between the UAVs and users. Therefore, the channel gains between UAV S/J and GR/Eve during flight slot n follow the free-space path loss model, which is shown as

$$h_{xy}[n]=\beta_0 d_{xy}^{-2}[n]=\frac{\beta_0}{\left\|\mathbf{q}_x[n]-\mathbf{w}_y\right\|^2}, x\in\{s,j\}, y\in\{g,e\}, \forall n \tag{3}$$

where β_0 is the channel power gain at reference distance. Define $P_s[n]$ and $P_j[n]$ are the UAV S/J's transmit power in slot n, respectively. Taking the average/peak power limits into account, the power constraint (4) of UAV S /J can be described as

$$\frac{1}{N}\sum_{n=1}^{N}P_x[n]\le\bar{P}_x, 0\le P_x[n]\le P_{x\max}, x\in\{s,j\}, \forall n \tag{4}$$

where $\bar{P}_x$ and $P_{x\max}$ are the average and peak transmit power, respectively. According to [29], a lower bound on the secrecy rate of the mobile jammer enabled secure UAV communication network in bits per channel use (BPCU) over the N equal-length time slots is given by

$$\begin{aligned} R[n]=&\left(\log_2\left(1+\gamma_g[n]\right)-\log_2\left(1+\gamma_e[n]\right)\right. \\ &\left.-\sqrt{\frac{V_g[n]}{L}}\frac{Q^{-1}(\varepsilon)}{\ln 2}-\sqrt{\frac{V_e[n]}{L}}\frac{Q^{-1}(\delta)}{\ln 2}\right)^+, \forall n \end{aligned} \tag{5}$$

where $(\chi)^+\triangleq\max(\chi,0)$, L is the packet length, ε is the packet error rate at GR and δ is the information leakage at Eve, $Q(x)=\frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-\frac{t^2}{2}}dt$ is the right tail function of standard normal distribution. $\gamma_g[n]$ and $\gamma_e[n]$ are the signal to interference plus noise radio (SINR) of GR and Eve, which can be shown as

$$\begin{aligned} \gamma_g[n]&=\frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_g\right\|^2}{\left(P_d[n]\zeta_0/\left\|\mathbf{q}_d[n]-\mathbf{w}_g\right\|^2\right)+1} \\ \gamma_e[n]&=\frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_e\right\|^2}{\left(P_d[n]\zeta_0/\left\|\mathbf{q}_d[n]-\mathbf{w}_e\right\|^2\right)+1} \end{aligned} \tag{6}$$

where $\zeta_0=\beta_0/\sigma^2$, σ^2 is the average noise power, and the channel dispersion of GR and Eve can be described as $V_y[n]=1-\left(1+\gamma_y[n]\right)^{-2}, y\in\{g,e\}, \forall n$.

Since the GR's packet error rate cannot be ignored, the AESR is considered to be maximized by jointly optimizing the UAVs' trajectories and power under the optimal packet error rate, and the optimization problem should satisfy the UAVs' mobility constraints (1) and the peak/average power constraints (4). Define $\mathbf{P}_x=\left\{P_x[n], x\in\{s,j\}, \forall n\right\}$ and $\mathbf{Q}_x=\left\{\mathbf{q}_x[n], x\in\{s,j\}, \forall n\right\}$ are the power and trajectories of UAV x in flight time slot n, respectively. Therefore, the resultant AESR maximization problem can be given by

$$\text{(OP)}: \max_{\mathbf{P}_s,\mathbf{P}_d,\mathbf{Q}_s,\mathbf{Q}_d,\varepsilon}\frac{1}{N}\sum_{n=1}^{N}R[n](1-\varepsilon) \tag{7a}$$

$$\text{s.t.}\quad \left\|\mathbf{q}_s[n+1]-\mathbf{q}_s[n]\right\|\le V_{\max}\delta_t, \forall n \tag{7b}$$

$$\left\|\mathbf{q}_j[n+1]-\mathbf{q}_j[n]\right\|\le V_{\max}\delta_t, \forall n \tag{7c}$$

$$\mathbf{q}_s[1]=\mathbf{q}_{sI}, \mathbf{q}_s[N]=\mathbf{s}_{jF} \tag{7d}$$

$$\mathbf{q}_j[1]=\mathbf{q}_{jI}, \mathbf{q}_j[N]=\mathbf{q}_{jF} \tag{7e}$$

$$\frac{1}{N}\sum_{n=1}^{N}P_s[n]\le\bar{P}_s, 0\le P_s[n]\le P_{s\max}, \forall n \tag{7f}$$

$$\frac{1}{N}\sum_{n=1}^{N}P_j[n]\le\bar{P}_j, 0\le P_j[n]\le P_{j\max}, \forall n \tag{7g}$$

$$0<\varepsilon<\varepsilon_{\max} \tag{7h}$$

where the operation $(\cdot)^+$ can be omitted because if there exists some flight time slot making $R[n]<0$, we will set $P_x[n]=0, x\in\{s,j\}, \forall n$, and then the value of $R[n]$ can be larger than zero while satisfying the peak/average power constraint (4). Besides, the constraint (7h) guarantees the reliable data transmission, and $\varepsilon_{\max}$ is the maximum packet error rate. It should be pointed that the optimization problem is still nonconvex and complicated to solve. On this basis, an efficient alternating optimization algorithm will be proposed to guarantee the convergence and effectiveness of AESR maximization problem in this paper, then we can find a locally optimal solution of the optimization problem (OP).

3. Optimal solutions for (OP)

Based on the above discussions, slack variables $\mathbf{T}_y=\left\{t_y[n], \forall n\right\}$ and $\mathbf{M}_y=\left\{m_y[n], \forall n\right\}$ can be introduced to facilitate processing the (OP). Mathematically, this optimization problem is reformulated as

$$\text{(OP1)}: \max_{\mathbf{P}_s,\mathbf{P}_d,\mathbf{Q}_s,\mathbf{Q}_d,\varepsilon}\frac{1}{N}\sum_{n=1}^{N}\bar{R}[n](1-\varepsilon) \tag{8a}$$

$$\text{s.t.}\quad \text{(7b)–(7h)} \tag{8b}$$

$$t_y[n]\ge\frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2}{\left(P_j[n]\zeta_0/\left\|\mathbf{q}_j[n]-\mathbf{w}_y\right\|^2\right)+1}, y\in\{g,e\}, \forall n \tag{8c}$$

$$m_y[t]\ge 1-\left(1+t_y[n]\right)^{-2}, y\in\{g,e\}, \forall n \tag{8d}$$

$$m_y[t]\ge 0, y\in\{g,e\}, \forall n \tag{8e}$$

with

$$\begin{aligned} \bar{R}[n]=&\log_2\left(1+\frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_g\right\|^2}{\left(P_d[n]\zeta_0/\left\|\mathbf{q}[n]-\mathbf{w}_g\right\|^2\right)+1}\right) \\ &-\log_2\left(1+t_e[n]\right)-m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}}-m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}} \end{aligned} \tag{9}$$

where constraint (8c) makes (OP) more tractable, and constraints (8d), (8e) can make the proposed alternating algorithm smooth running. Furthermore, constraints (8c) and (8d) can ensure (OP1) is equivalent to (OP) if they keep with equality at the maximal AESR, and the AESR can be increased by decreasing $\mathbf{T}_y$ and $\mathbf{M}_y$. Especially, (OP1) is a nonconvex problem which is complex to address since the optimization function (8a) is non-concave in regard to $\mathbf{P}_x$, $\mathbf{Q}_x$ and $\mathbf{T}_e$, and constraints (8c) and (8d) are still non-convex with respect to the UAVs' power and trajectories in each flight time slot. On this basis, it can verify that there exists a unique optimal packet error rate to maximize the AESR.

Algorithm 1 : SCA algorithm for AESR maximization problem

1: **Initialization**: $\left\{\mathbf{P}_x^{(0)}, \mathbf{Q}_x{}^{(0)}, \mathbf{T}_y^{(0)}, \mathbf{M}_y{}^{(0)}\right\}, x \in \{s, j\}, y \in \{g, e\}$ in $\bar{R}$, algorithm accuracy τ, choose a step size $\alpha \in (0, 1]$, and set the iteration k=0;
2: **repeat**
3: Find the optimal value $m^{(k)}$ to (OPt) by CVX,t $\in \{2b, 3b, 4b, 5b\}$, and compute $\bar{R}^{(k)}$.
4: Set $k_{\text{opt}} = k$.
5: Compute $m_{fea}^{(k)}$ and the value of $\bar{R}\left(m_{fea}^{(k)}\right)$.
6: Set $m^{(k+1)} = m^{(k)} + \alpha\left(m_{fea}^{(k)} - m^{(k)}\right)$.
7: Set $k \leftarrow k + 1$
8: **until** the fraction increase of the optimization problem is below a threshold τ.

As a result, an efficient alternating optimization algorithm based on Bisection method and SCA technique will be proposed to solve the AESR maximization problem. Specifically, the (OP) can be divided into five subproblems to maximize the UAV S/J's power $\mathbf{P}_s/\mathbf{P}_j$, the trajectory $\mathbf{Q}_s/\mathbf{Q}_j$ and the packet error rate in an alternately iterative manner.

3.1. Optimization of the source UAV's transmit power $\mathbf{P}_s$

For given feasible packet error rate ε, UAV J's power $\mathbf{P}_j$, UAV S/J's trajectories $\mathbf{Q}_s$ and $\mathbf{Q}_j$, (OP1) can be simplified as

$$(\text{OP2a}): \max_{\mathbf{P}_s, \mathbf{T}_y, \mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N} \bar{R}[n](1-\varepsilon) \tag{10a}$$

$$\text{s.t.} \ \frac{1}{N}\sum_{n=1}^{N} P_s[n] \le \bar{P}_s, 0 \le P_s[n] \le P_{s\max}, \forall n \tag{10b}$$

$$t_y[n] \ge \frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2}{\left(P_j[n]\zeta_0/\left\|\mathbf{q}_j[n]-\mathbf{w}_y\right\|^2\right)+1}, y \in \{g, e\}, \forall n \tag{10c}$$

$$m_y[t] \ge 1 - \left(1 + t_y[n]\right)^{-2}, y \in \{g, e\}, \forall n \tag{10d}$$

$$m_y[t] \ge 0, y \in \{g, e\}, \forall n \tag{10e}$$

Obviously, optimization problem (OP2a) is still non-convex since the second term of $\bar{R}[n]$ is non-concave and the constraint (10d) is non-convex. Then, we concentrate on converting (OP2a) into a convex AESR maximization problem. When the AESR reaches the optimal solution, it can be obtained that the optimization problem (OP2a) is equivalent to (OP1) and constraints (10c) and (10d) keep with equality.

Constraint (10d) is non-convex now since it is the superlevel of a standard convex function. Consequently, we can utilize the first-order Taylor expansion to transform it into a convex constraint. Given initial feasible points $\mathrm{t}_{y-\text{fea}}[n]$ and $\mathrm{m}_{y-\text{fea}}[n]$, $y \in \{g, e\}, \forall n$, constraint (10d) can be reformulated as

$$\begin{aligned} &t_{y-\text{ fea}}^2[n] + 2t_{y-\text{ fea}}[n]\left(t_y[n] - t_{y\text{-fea}}[n]\right) \ge 1 - \left(1 + m_{y-\text{ fea}}[n]\right)^{-2} \\ &+ 2\left(1 + m_{y-\text{ fea}}[n]\right)^{-3}\left(m_y[n] - m_{y\text{-fea}}[n]\right), y \in \{g, e\}, \forall n \end{aligned} \tag{11}$$

For the optimization problem (10a), by employing the first-order Taylor expansion, the second term of $\bar{R}[n]$ can be reconstructed by its convex lower bound, which is given as

$$-\log_2\left(1 + t_e[n]\right) \ge -\left(\log_2\left(1 + t_{e-\beta}a[n]\right) + \frac{t_e[n] - t_{e-\beta a}[n]}{\ln 2\left(1 + t_{e-fea}[n]\right)}\right) \tag{12}$$

Define $A_{gn} = \frac{\zeta_0/\|\mathbf{q}_s[n]-\mathbf{w}_g\|^2}{\left(\zeta_0 P_j[n]/\left\|\mathbf{q}_j[n]-\mathbf{w}_g\right\|^2\right)+1}$, then the optimization problem (OP2a) is approximated as

$$(\text{OP2b}): \max_{\mathbf{P}_s, \mathbf{T}_y, \mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N} \bar{R}_{ps}^{lb}[n](1-\varepsilon) \tag{13a}$$

$$\text{s.t. (10b), (10c), (10e), (11)} \tag{13b}$$

with

$$\begin{aligned} &\bar{R}[n] \ge \bar{R}_{ps}^{lb}[n] = \log_2\left(1 + A_{gn}[n]P_s[n]\right) - \log_2\left(1 + t_{e-fea}[n]\right) \\ &- \frac{t_e[n] - t_{e-\text{ fea }}[n]}{\ln 2\left(1 + t_{e-\text{ fea }}[n]\right)} - m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}} \end{aligned} \tag{14}$$

After such approximation, subproblem (OP2b) has been transformed into a standard convex optimization problem, which can be optimally solved by CVX [30]. Therefore, the power optimization problem of source UAV can be obtained by Algorithm 1.

3.2. Optimization of the jammer UAV's transmit power $\mathbf{P}_j$

For given feasible packet error rate ε, UAV S's power $\mathbf{P}_s$, UAV S/J's trajectories $\mathbf{Q}_s$ and $\mathbf{Q}_j$, (OP1) can be rewritten as

$$(\text{OP3a}): \max_{\mathbf{P}_j, \mathbf{T}_y, \mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N} \bar{R}[n](1-\varepsilon) \tag{15a}$$

$$\text{s.t.} \ \frac{1}{N}\sum_{n=1}^{N} P_j[n] \le \bar{P}_j, 0 \le P_j[n] \le P_{j\max}, \forall n \tag{15b}$$

$$t_y[n] \ge \frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2}{\left(P_j[n]\zeta_0/\left\|\mathbf{q}_j[n]-\mathbf{w}_y\right\|^2\right)+1}, y \in \{g, e\}, \forall n \tag{15c}$$

$$m_y[t] \ge 1 - \left(1 + t_y[n]\right)^{-2}, y \in \{g, e\}, \forall n \tag{15d}$$

$$m_y[t] \ge 0, y \in \{g, e\}, \forall n \tag{15e}$$

Noting that optimization problem (OP3a) is equivalent to (OP1) when constraints (15c) and (15d) keep with equality, and this can make AESR obtain the optimal solution. Define $B_{gn1} = P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_g\right\|^2$, $B_{gn2} = \zeta_0/\left\|\mathbf{q}_j[n]-\mathbf{w}_g\right\|^2$, then AESR can be expressed as

$$\begin{aligned} \bar{R}[n] &= \log_2\left(1 + \frac{B_{gnl}}{P_j[n]B_{gn2}+1}\right) - \log_2\left(1 + t_e[n]\right) \\ &- m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}} \end{aligned} \tag{16}$$

Obviously, due to the non-convexity of the constraint (15d) and the convexity of the first term of objective function $\bar{R}[n]$, (OP3a) is a non-convex optimization problem, so we use the first-order Taylor expansion to obtain the lower bound of $\log_2\left(1 + \frac{B_{gn1}}{P_j[n]B_{gn2}+1}\right)$. On this basis, for given feasible points $P_{j-\text{ fea }}[n]$, we can obtain

$$\begin{aligned} &\log_2\left(1 + \frac{B_{gn1}}{P_j[n]B_{gn2}+1}\right) \ge \log_2\left(1 + \frac{B_{gn1}}{P_{j-fea}[n]B_{gn2}+1}\right) \\ &- \frac{B_{gn1}B_{gn2}\left(P_j[n] - P_{j-fea}[n]\right)}{\ln 2\left(B_{gn1} + B_{gn2}P_{j-fea}[n]+1\right)\left(B_{gn2}P_{j-fea}[n]+1\right)} \end{aligned} \tag{17}$$

Similarly, constraint (15d) can also be given by constraint (11). Therefore, the problem (OP3b) is given by

$$(\text{OP3b}): \max_{\mathbf{P}_j, \mathbf{T}_y, \mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N} \bar{R}_{pj}^{lb}[n](1-\varepsilon) \tag{18a}$$

$$\text{s.t. (11), (15b), (15c), (15e)} \tag{18b}$$

with

$$\begin{aligned} \bar{R}[n] \ge \bar{R}_{pj}^{lb}[n] &= \log_2\left(1 + \frac{B_{gn1}}{P_{j-fea}[n]B_{gn2}+1}\right) \\ &- \frac{B_{gn1}B_{gn2}\left(P_j[n] - P_{j-fea}[n]\right)}{\ln 2\left(B_{gn1} + B_{gn2}P_{j-fea}[n]+1\right)\left(B_{gn2}P_{j-fea}[n]+1\right)} \end{aligned}$$

$$- \log_2\left(1+t_{e-fea}[n]\right) - \frac{\left(t_e[n]-t_{e-fea}[n]\right)}{\ln 2\left(1+t_{e-fea}[n]\right)} - m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}} \tag{19}$$

Based on the above analysis, (OP3b) has been transformed into a standard convex optimization problem, which can be efficiently solved by CVX. Therefore, the power optimization problem of jammer UAV can also be obtained by Algorithm 1.

3.3. Optimization of the source UAV's trajectory $\mathbf{Q}_s$

For given feasible packet error rate ε, UAV J's trajectory $\mathbf{Q}_j$, UAV S/J's power $\mathbf{P}_s$ and $\mathbf{P}_j$, (OP1) can be rewritten as

$$(\text{OP4a}): \max_{\mathbf{Q}_s,\mathbf{T}_y,\mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N}\bar{R}[n](1-\varepsilon) \tag{20a}$$

$$\text{s.t.}\ \left\|\mathbf{q}_s[n+1]-\mathbf{q}_s[n]\right\| \le V_{\max}\delta_t, n=1,\dots,N-1 \tag{20b}$$

$$\mathbf{q}_s[1]=\mathbf{q}_{sI}, \mathbf{q}_s[N]=\mathbf{q}_{sF} \tag{20c}$$

$$t_y[n] \ge \frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2}{\left(P_j[n]\zeta_0/\left\|\mathbf{q}_j[n]-\mathbf{w}_y\right\|^2\right)+1}, y\in\{g,e\}, \forall n \tag{20d}$$

$$m_y[t] \ge 1-\left(1+t_y[n]\right)^{-2}, y\in\{g,e\}, \forall n \tag{20e}$$

$$m_y[t] \ge 0, y\in\{g,e\}, \forall n \tag{20f}$$

It is noteworthy that constraints (20d) and (20e) can be satisfied with equality at the optimal solution of problem (OP4a), conversely objective value can be increased by decreasing $t_y[n]$ and $m_y[n]$. Define $C_g = \frac{\zeta_0 P_s[n]}{\left(\zeta_0 P_j[n]/\left\|\mathbf{q}_j[n]-\mathbf{w}_g\right\|^2\right)+1}, \forall n$, $\bar{R}[n]$ is simplified as

$$\bar{R}[n] = \log_2\left(1+\frac{C_g}{\left\|\mathbf{q}_s[n]-\mathbf{w}_g\right\|}\right) - \log_2\left(1+t_e[n]\right) - m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}} \tag{21}$$

Obviously, constraints (20d) and (20e) are non-convex such that we can transform them into convex constraints by SCA technique. On this basis, slack variable $\mathbf{L}_y = \left\{l_y[n], y\in\{g,e\}, \forall n\right\}$ should be introduced and constraint (20d) can be described as

$$t_y[n] \ge \frac{C_y}{l_y[n]}, y\in\{g,e\}, \forall n \tag{22a}$$

$$\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2 \ge l_y[n], y\in\{g,e\}, \forall n \tag{22b}$$

where Eq. (22a) is a convex constraint. Since the term $\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2$ of Eq. (22b) is the lower bound of a convex function, then we can transform it into a convex constraint by using the first-order Taylor expansion. Therefore, for given initial feasible points $\mathbf{q}_{s-fea}[n]$, Eq. (22b) can be expressed as

$$\left\|\mathbf{q}_{s-fea}[n]-\mathbf{w}_y\right\|^2 + 2\left(\mathbf{q}_{s-fea}[n]-\mathbf{w}_y\right)^T\left(\mathbf{q}_s[n]-\mathbf{q}_{s-fea}[n]\right) \ge l_y[n], y\in\{g,e\}, \forall n \tag{23}$$

where Eq. (23) is a convex constraint now.

For Eq. (21), we can obtain that $\bar{R}[n]$ is a convex function with respect to $\left\|\mathbf{q}_s[n]-\mathbf{w}_g\right\|^2$ and $t_e[n]$, which can utilize first-order approximation technique to solve the nonconcave problem. Consequently, for the given feasible point $\left\|\mathbf{q}_{s-fea}[n]-\mathbf{w}_g\right\|^2$ and $t_{e-fea}[n]$, we obtain

$$\begin{aligned}\bar{R}[n] \ge \bar{R}^{lb}_{qs}[n] &= \log_2\left(1+\frac{C_g}{\left\|\mathbf{q}_{s-\text{ fea }}[n]-\mathbf{w}_g\right\|^2}\right) \\ &- \frac{C_g\left(\left\|q_s[n]-\mathbf{w}_g\right\|^2-\left\|\mathbf{q}_{s-fea}[n]-\mathbf{w}_g\right\|^2\right)}{\ln 2\left\|\mathbf{q}_{s-fea}[n]-\mathbf{w}_g\right\|^2\left(\left\|\mathbf{q}_{s-fea}[n]-\mathbf{w}_g\right\|^2+C_g\right)} \\ &- \log_2\left(1+t_{e-fea}[n]\right) - \frac{\left(t_e[n]-t_{e-fea}[n]\right)}{\ln 2\left(1+t_{e-fea}[n]\right)} \\ &- m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}}\end{aligned} \tag{24}$$

Analytically, optimization problem (OP4b) can be described as

$$(\text{OP4b}): \max_{\mathbf{Q}_s,\mathbf{T}_y,\mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N}\bar{R}^{lb}_{qs}[n](1-\varepsilon) \tag{25a}$$

$$\text{s.t.}\ (11), (20b), (20c), (20f), (22a), (23) \tag{25b}$$

Based on the above analysis, the objective function of (OP4b) is concave and the constraints are convex now such that the CVX can be utilized to solve it optimally. Therefore, the trajectory optimization problem of source UAV can be similarly obtained by Algorithm 1.

3.4. Optimization of the jammer UAV's trajectory $\mathbf{Q}_j$

For given feasible packet error rate ε, UAV S's trajectory $\mathbf{Q}_s$, UAV S/J's power $\mathbf{P}_s$ and $\mathbf{P}_j$, (OP1) can be rewritten as [4]

$$(\text{OP5a}): \max_{\mathbf{Q}_j,\mathbf{T}_y,\mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N}\bar{R}[n](1-\varepsilon) \tag{26a}$$

$$\text{s.t.}\ \left\|\mathbf{q}_j[n+1]-\mathbf{q}_j[n]\right\| \le V_{\max}\delta_t, n=1,\dots,N-1 \tag{26b}$$

$$\mathbf{q}_j[1]=\mathbf{q}_{jI}, \mathbf{q}_j[N]=\mathbf{q}_{jF} \tag{26c}$$

$$t_y[n] \ge \frac{P_s[n]\zeta_0/\left\|\mathbf{q}_s[n]-\mathbf{w}_y\right\|^2}{\left(P_j[n]\zeta_0/\left\|\mathbf{q}_j[n]-\mathbf{w}_y\right\|^2\right)+1}, y\in\{g,e\}, \forall n \tag{26d}$$

$$m_y[t] \ge 1-\left(1+t_y[n]\right)^{-2}, y\in\{g,e\}, \forall n \tag{26e}$$

$$m_y[t] \ge 0, y\in\{g,e\}, \forall n \tag{26f}$$

Constraints (26d) and (26e) can be satisfied with equality at the optimal solution of problem (OP5a), conversely objective value can be increased by decreasing $t_y[n]$ and $m_y[n]$. Define $N_{e1} = P_s[n]\frac{\zeta_0}{\left\|\mathbf{q}_s[n]-\mathbf{w}_e\right\|^2}$, $N_{e2} = \zeta_0 P_j[n], \forall n$, then objective function $\bar{R}[n]$ can be simplified as

$$\bar{R}[n] = \log_2\left(1+t_g[n]\right) - \log_2\left(1+\frac{N_{e1}}{\left(N_{e2}/\left\|\mathbf{q}_{j-fea}[n]-\mathbf{w}_e\right\|^2\right)+1}\right) - m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}} \tag{27}$$

It can be verified that the second term of Eq. (27) is non-concave that we can apply SCA technique to transform $\bar{R}[n]$ into a concave function. Similarly, by using the first-order Taylor expansion for the feasible point $\left\|q_{j-fea}[n]-\mathbf{w}_e\right\|^2$, we have

$$\bar{R}[n] \ge \bar{R}^{lb}_{qj}[n] = \log_2\left(1+t_g[n]\right) - \log_2\left(1+\frac{N_{e1}}{\left(N_{ge}/\left\|\mathbf{q}_{j-fea}[n]-\mathbf{w}_e\right\|^2\right)+1}\right)$$

Algorithm 2 : Alternating optimization algorithm for AESR problem

1: **Initialization**: Find an initial feasible solution $\mathbf{P}_s^{(0)}, \mathbf{P}_j^{(0)}, \mathbf{q}_s^{(0)}, \mathbf{q}_j^{(0)}, \varepsilon^{(0)}$ and the AESR is $\bar{R}^{(0)}$; Define maximum iterations k_{max}, algorithm accuracy μ, and let $k = 0$;
2: **While** $k \le k_{\max}$ or $\tau > \mu$
3: For given $\mathbf{P}_j^{(k)}, \mathbf{q}_s^{(k)}, \mathbf{q}_j^{(k)}$ and $\varepsilon^{(k)}$, set the feasible solution $\mathrm{P}_{s-fea} = \mathbf{P}_s^{(k)}$, then update the optimal source UAV's power $\mathbf{P}_s^{(k+1)}$ by applying Algorithm 1.
4: For given $\mathbf{P}_s^{(k+1)}, \mathbf{q}_s^{(k)}, \mathbf{q}_j^{(k)}$ and $\varepsilon^{(k)}$, set the feasible solution $\mathrm{P}_{j-fea} = \mathbf{P}_j^{(k)}$, then update the optimal source UAV's power $\mathbf{P}_j^{(k+1)}$ by applying Algorithm 1.
5: For given $\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_j^{(k)}$ and $\varepsilon^{(k)}$, set the feasible solution $\mathrm{q}_{s-fea} = \mathbf{P}_s^{(k)}$, then update the optimal source UAV's power $\mathbf{q}_s^{(k+1)}$ by applying Algorithm 1.
6: For given $\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}$ and $\varepsilon^{(k)}$, set the feasible solution $\mathrm{q}_{j-fea} = \mathbf{P}_j^{(k)}$, then update the optimal source UAV's power $\mathbf{q}_j^{(k+1)}$ by applying Algorithm 1.
7: For given $\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}$ and $\mathbf{q}_j^{(k+1)}$, update the optimal packet error rate $\varepsilon^{(k+1)}$ by applying Bisection method.
8: Update $\bar{R}^{(k+1)}$, calculate $\tau = \left(\bar{R}^{(k+1)} - \bar{R}^{(k)}\right)/\bar{R}^{(k)}$
9: Update $k = k + 1$
10: **end While**
11: **Output:** $\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}, \mathbf{q}_j^{(k+1)}, \boldsymbol{\varepsilon}^{(k+1)}, \bar{R}^{(k+1)}$

$$
\begin{aligned}
&- \frac{N_{e1} N_{e2}\left(q_{je} - q_{je-fea}\right)}{\ln 2\left(q_{je-fea} + N_{e1} q_{je-fea} + N_{e2}\right)\left(N_{e2} + q_{je-fea}\right)} \\
&- m_g[n]\frac{Q^{-1}(\varepsilon)}{\ln 2\sqrt{L}} - m_e[n]\frac{Q^{-1}(\delta)}{\ln 2\sqrt{L}}
\end{aligned} \tag{28}
$$

where $q_{je} = \left\|\mathbf{q}_j[n] - \mathbf{w}_e\right\|^2$, $q_{je-fea} = \left\|\mathbf{q}_{j-fea}[n] - \mathbf{w}_e\right\|^2$. Therefore, optimization problem (OP5b) can be formulated as

$$
(\text{OP5b}): \max_{\mathbf{Q}_j, \mathbf{T}_y, \mathbf{M}_y} \frac{1}{N}\sum_{n=1}^{N} \bar{R}_{qj}^{lb}[n](1-\varepsilon) \tag{29a}
$$

$$
\text{s.t. } (11), (26b), (26c), (26f), (22a), (23) \tag{29b}
$$

At this point, problem (OP5b) has been transformed into a standard convex optimization problem, which can be solved by CVX. Therefore, the trajectory optimization problem of jammer UAV can be obtained similarly by Algorithm 1.

3.5. Optimization of packet error rate ε

For given feasible UAV S/J's transmit power and trajectories, the packet error rate is identical in each time slot in this scenario, i.e., $\frac{1}{N}\sum_{n=1}^{N}\varepsilon[n] = \varepsilon$. Define $Z_1[n] = \frac{\sqrt{V_g[n]}}{\ln 2\sqrt{L}}$, $Z_2[n] = \log_2\left(1+\gamma_g[n]\right) - \log_2\left(1+\gamma_e[n]\right) - \sqrt{\frac{V_e[n]}{L}}\frac{Q^{-1}(\delta)}{\ln 2}$. Therefore, (OP1) can be expressed as

$$
(\text{OP6}): \max_{\varepsilon} \frac{1}{N}\sum_{n=1}^{N} \bar{R}[n](1-\varepsilon) \tag{30a}
$$

$$
\text{s.t. } \check{\varepsilon} < \varepsilon < \varepsilon_{\max} \tag{30b}
$$

where $\bar{R}[n] = Z_2[n] - Z_1[n]Q^{-1}(\varepsilon)$ and $\check{\varepsilon} = Q\left(\ln 2 Z_2[n]/Z_1[n]\right)$. On this basis, we consider $\check{\varepsilon} < \varepsilon_{\max}$ to ensure the effectiveness of the optimal solution of (OP6). The value of AESR with respect to ε can be obtained as

$$
\begin{aligned}
\tilde{R}(\varepsilon) &= (1-\varepsilon)\frac{1}{N}\sum_{n=1}^{N} R[n] = (1-\varepsilon)\frac{1}{N}\sum_{n=1}^{N}\left(Z_2[n] - Z_1[n]Q^{-1}(\varepsilon)\right) \\
&= (1-\varepsilon)\left(\frac{1}{N}\sum_{n=1}^{N}\left(Z_2[n]\right) - Q^{-1}(\varepsilon)\frac{1}{N}\sum_{n=1}^{N}\left(Z_1[n]\right)\right) \\
&= (1-\varepsilon)\left(Z_{3n} - Q^{-1}(\varepsilon)Z_{4n}\right)
\end{aligned} \tag{31}
$$

where $Z_{3n} = \frac{1}{N}\sum_{n=1}^{N}\left(Z_2[n]\right), Z_{4n} = \frac{1}{N}\sum_{n=1}^{N}\left(Z_1[n]\right)$. Therefore, the first-order derivation with respect to ε can be given as

$$
\tilde{R}'(\varepsilon) = \left(Z_{4n}Q^{-1}(\varepsilon) - Z_{3n}\right) - (1-\varepsilon)Z_{4n}\left(Q^{-1}(\varepsilon)\right)' \tag{32}
$$

and the second-order derivative is

$$
\tilde{R}''(\varepsilon) = \left(-Z_{4n}\left(Q^{-1}(\varepsilon)\right)''\right)(1-\varepsilon) + 2Z_{4n}\left(Q^{-1}(\varepsilon)\right)' \tag{33}
$$

where $\left(Q^{-1}(\varepsilon)\right)' = -\sqrt{2\pi}\exp\left(\left(Q^{-1}(\varepsilon)\right)^2/2\right), \left(Q^{-1}(\varepsilon)\right)'' = -2\pi Q^{-1}(\varepsilon)\exp\left(\left(Q^{-1}(\varepsilon)\right)^2\right)$. Note that, when $\check{\varepsilon} < \varepsilon < \varepsilon_{\max}$, we can obtain that $\left(Q^{-1}(\varepsilon)\right)' \le 0$ and $\left(Q^{-1}(\varepsilon)\right)'' \ge 0$. Therefore, it can be derived that $\lim_{\varepsilon\to\check{\varepsilon}}\tilde{R}'(\varepsilon) \to +\infty$ and $\lim_{\varepsilon\to\varepsilon_{\max}}\tilde{R}'(\varepsilon) \to 0$. On this basis, it is also proved that $\tilde{R}''(\varepsilon) \le 0$, thus, $\tilde{R}(\varepsilon)$ is a concave function with respect to ε. Based on the above analysis, there exists a unique value of packet error rate that can maximize the AESR, and we can utilize the Bisection method [31] to obtain the optimal packet error rate ε^* for any given transmit power and trajectories of UAVs. The search interval of this algorithm is $\left[\check{\varepsilon}, \varepsilon_{\max}\right]$, algorithm accuracy is ρ, and the corresponding complexity can be shown as $\mathcal{O}\left(\log_2 n\right)$, where $n = \left(\varepsilon_{\max} - \check{\varepsilon}\right)/\rho$.

3.6. Proposed iterative algorithm

Based on the above analysis, we have transformed the UAV S/J's transmit power optimization problem and trajectory optimization problem into the standard convex optimization problems, and it has been verified that there exists a unique optimal packet error rate ε^* to maximize the AESR. On this basis, an efficient iterative algorithm is proposed by jointly optimizing subproblem (OP2b), (OP3b), (OP4b), (OP5b) and (OP6) in an iterative manner. The detailed algorithm is summarized in Algorithm 2.

According to [32], $\bar{R}[n]$ is a non-decreasing function at each iteration, and the convergence of proposed algorithm is proved as follows

$$
\begin{aligned}
&\bar{R}\left(\mathbf{P}_s^{(k)}, \mathbf{P}_j^{(k)}, \mathbf{q}_s^{(k)}, \mathbf{q}_d^{(k)}, \varepsilon^{(k)}\right) \\
&\overset{(a)}{\le} \bar{R}\left(\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k)}, \mathbf{q}_s^{(k)}, \mathbf{q}_j^{(k)}, \varepsilon^{(k)}\right) \\
&\le \bar{R}\left(\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k)}, \mathbf{q}_j^{(k)}, \varepsilon^{(k)}\right) \\
&\overset{(c)}{\le} \bar{R}\left(\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}, \mathbf{q}_j^{(k)}, \varepsilon^{(k)}\right) \\
&\overset{(d)}{\le} \bar{R}\left(\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}, \mathbf{q}_j^{(k+1)}, \varepsilon^{(k)}\right) \\
&\overset{(e)}{\le} \bar{R}\left(\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}, \mathbf{q}_j^{(k+1)}, \varepsilon^{(k+1)}\right)
\end{aligned} \tag{34}
$$

where conditions (a), (b), (c), (d) and (e) hold due to the fact that $\mathbf{P}_s^{(k+1)}, \mathbf{P}_j^{(k+1)}, \mathbf{q}_s^{(k+1)}, \mathbf{q}_j^{(k+1)}$ and $\varepsilon^{(k+1)}$ are the optimal solutions of subproblem (OP2b), (OP3b), (OP4b), (OP5b) and (OP6), respectively. Assuming that the maximum iterations of Algorithm 2 is k_{max}, variables $\mathbf{P}_s, \mathbf{P}_j, \mathbf{q}_s, \mathbf{q}_j$ and ε are convergent when $\bar{R}$ converge. Since Algorithm 1 with polynomial complexity $\mathcal{O}\left(N^{3.5}\right)$ and Bisection method with complexity $\mathcal{O}\left(\log_2 n\right)$ are needed to be addressed at each iteration, the complexity of Algorithm 2 can be described as $\mathcal{O}\left(k_{\max}\left(N^{3.5} + \log_2 n\right)\right)$. Above all, the proposed algorithm can utilize finite iterations to obtain the convergent solution of the AESR maximization problem.

Table 1
Parameters in numerical analysis.

Parameter	Value	Parameter	Value
$v_{\max}/(\mathrm{m/s})$	15	H/m	100
$\bar{P}_s/\mathrm{W}$	10^{-2}	$P_{s\max}/\mathrm{W}$	4×10^{-2}
$\bar{P}_j/\mathrm{W}$	10^{-3}	$P_{j\max}/\mathrm{W}$	4×10^{-3}
$\mathbf{w}_g/\mathrm{m}$	$[200, 100, 0]^T$	$\mathbf{w}_e/\mathrm{m}$	$[200, -100, 0]^T$
$\mathbf{q}_s[1]/\mathrm{m}$	$[0, 0, H]^T$	$\mathbf{q}_j[1]/\mathrm{m}$	$[0, 0, H]^T$
$\mathbf{q}_s[N]/\mathrm{m}$	$[400, 0, H]^T$	$\mathbf{q}_j[N]/\mathrm{m}$	$[400, 0, H]^T$
δ_t/s	1	L	800
μ	10^{-3}	δ	10^{-2}
ς_0/dB	80	ρ	10^{-3}

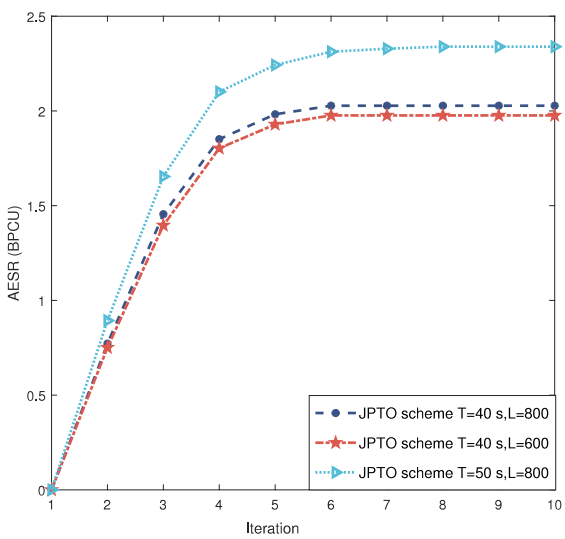


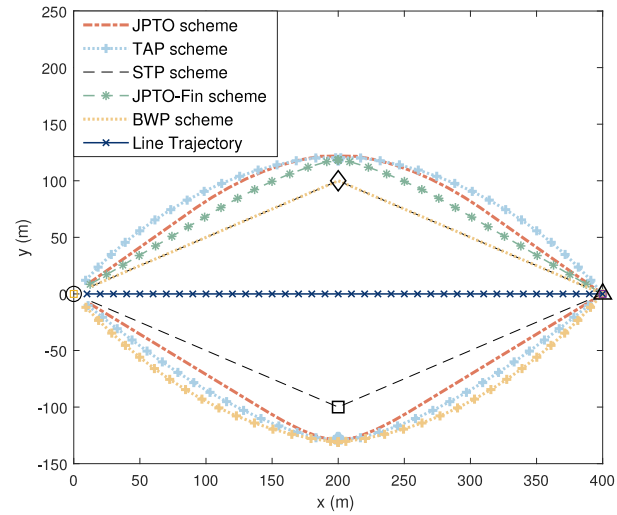
Fig. 2. AESR versus iterations for different values of the flight period and packet length.

4. Numerical results

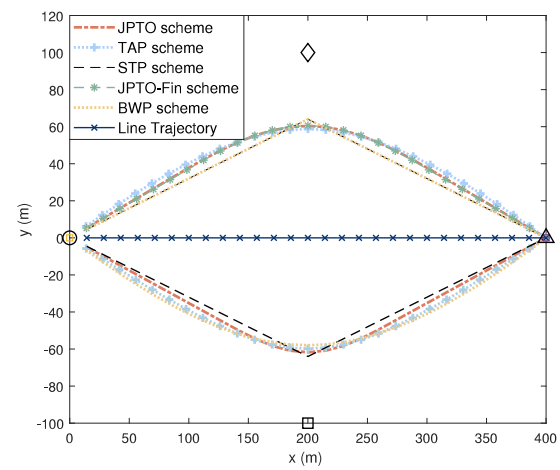
In this section, experimental results are provided to demonstrate the effectiveness of the proposed alternating optimization algorithm, i.e., the joint optimization of the UAVs' transmit power and trajectories under the optimal packet error rate (denoted as JPTO). The specific simulation parameters can be set up as Table 1. On this basis, the proposed JPTO scheme is compared with four different schemes. Scheme 1 is traj. & avg. pow scheme in Ref. [33], which is called TAP scheme in this paper. Scheme 2 is STP scheme which is given in Ref. [17], scheme 3 is JTPO-Fin scheme which is shown in Ref. [28], and scheme 4 is BET-Without-PC scheme in Ref. [25], which is expressed as BWP scheme.

Fig. 2 shows the AESR of the proposed alternating optimization algorithm versus iterations with different flight period and packet length. The flight height and maximum flight speed of UAV are $H = 100$ m and $v_{\max} = 15$ m/s, respectively. In addition, the accuracy of the proposed algorithm is 10^{-3}, the maximum iterations is 10^3, and the algorithm converges when the iterations $k \geq k_{\max}$ or $\left|\bar{R}^{(k+1)} - \bar{R}^{(k)}\right| < \mu$. It is observed that the AESR tends to be stable when the number of iterations is about 6. In other words, the proposed alternating algorithm is considerable efficient since it converges only in a few iterations. On this basis, flight period and packet length also have a significant impact on AESR. The longer the packet length, the larger the AESR. Besides, when the flight period becomes larger, it will have a longer hovering time that can communicate well with GR, resulting in a better AESR.

Fig. 3 shows the UAVs' trajectories obtained by the JPTO and other four benchmark schemes under different values of flight period. The UAV S/J's initial and final positions, as well as the positions of the GR and Eve are marked with ○, Δ, ⋄, and □ respectively. Before optimization, we define the initial flight line of UAV S as the line trajectory,



(a)



(b)

Fig. 3. UAVs' trajectories versus the flight period T under different schemes, where $T = 40$ s for (a), and $T = 28$ s for (b).

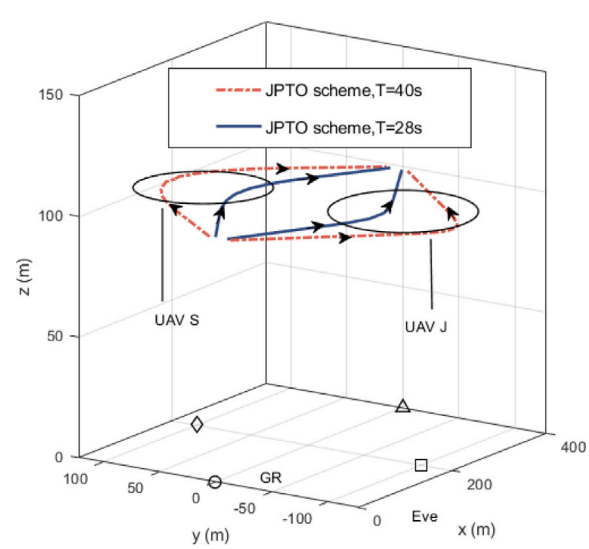
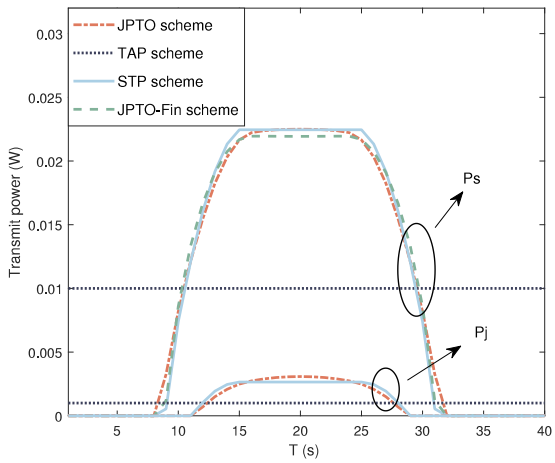
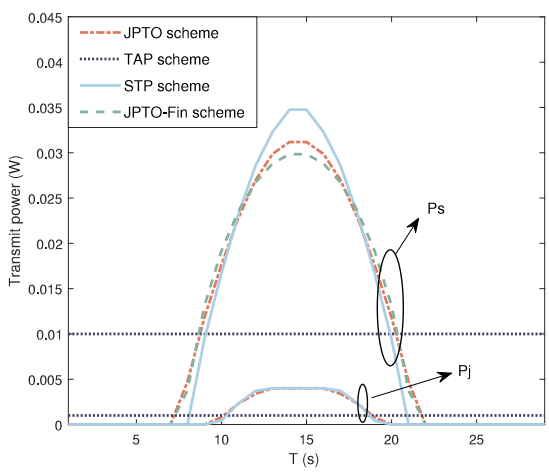


Fig. 4. UAVs' trajectories for different values of the flight period.



(a)



(b)

Fig. 5. UAVs' transmit power versus the flight period T under different schemes, where $T = 40$ s for (a), and $T = 29$ s for (b).

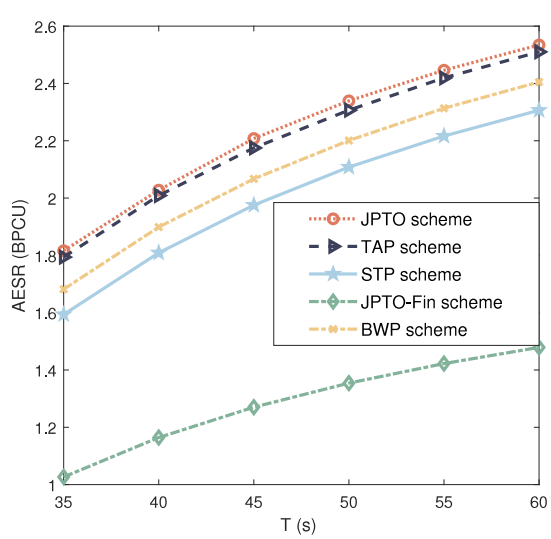


Fig. 6. AESR versus the flight period T under different schemes.

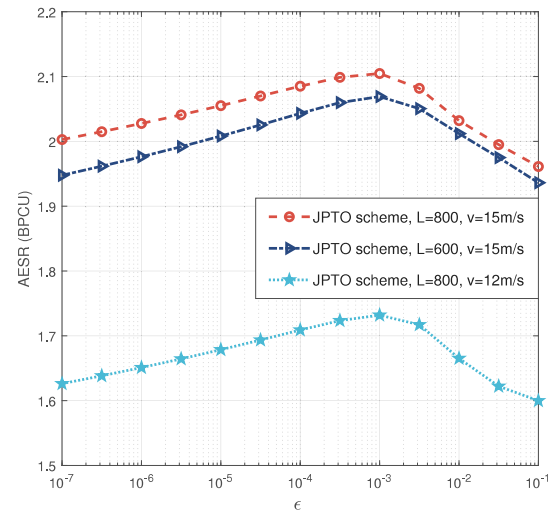


Fig. 7. AESR versus the packet error rate ϵ under different packet length and flight speed.

which is a straight line between the initial position and final position. From Fig. 3(a), it is shown that the trajectories of UAV S/J are identical for all schemes. Note that the UAV J flies in the opposite direction to UAV S, which can effectively deteriorate with Eve and obtain better security performance. This indicates that, mobile jammer UAV can effectively help UAV S realize better communication with GR. It can be obviously obtained that the trajectories obtained by the JPTO scheme and the benchmark TAP scheme are similar for different value of flight period T, i.e., as T increases, UAV S/J will fly as close as possible to the GR/Eve, respectively. Compared with STP scheme and BWP scheme, the source UAV's trajectory obtained by the proposed JPTO scheme and JTPO-Fin scheme can fly farther away from Eve, which can interfere with the Eve's wiretapping and improve the security performance. In addition, it is noteworthy that the flight trajectories of UAV S under STP scheme and BWP scheme are identical, while the trajectories of UAV J are distinct. This is indicated that UAV J can fly away from Eve to guarantee better secure communications. Moreover, it is found that the hovering location of UAV S/J is not directly above GR/Eve, this can be explained that the stationary location is quite significant for the tradeoff between communication performance and security performance. In the above case, the optimal stationary location can achieve the maximum AESR. From Fig. 3(b), the flying period is not enough for reaching the GR, thus the UAVs should fly to the final position in an arc path at the maximum speed to ensure communication performance within T, and the UAVs' trajectories for all schemes are identical. However, when T=40 s, the UAV S/J firstly fly to a certain position around GR/Eve at the maximum speed, then hovering the stationary location as long as possible before flying to the final position, which can improve the communication quality of legitimate channel and reduce information leakage. This observation can be demonstrated in Fig. 4, and the UAV S/J should fly as close as possible to the location of GR/Eve to guarantee the higher communication quality.

Fig. 5 shows the UAVs' transmit power of different schemes versus flying period T. The flight period of Fig. 5(a) and Fig. 5(b) are 40 and 29, respectively. Due to their symmetry, we can investigate the simulation results of the first half of the slots. From Fig. 5(a), we can obverse that P_j and P_s are essentially identical, firstly, UAV S/J's power increases when it flies from initial location to the GR/Eve, then the power keep unchanged for hovering, and finally the power will decrease to 0 when it flies to the final location. On this basis, if the flight period is not long enough to ensure hovering, the UAV S/J will fly to GR/Eve with higher power, then they will fly straightly to the final location without hovering. Note that the proposed schemes start

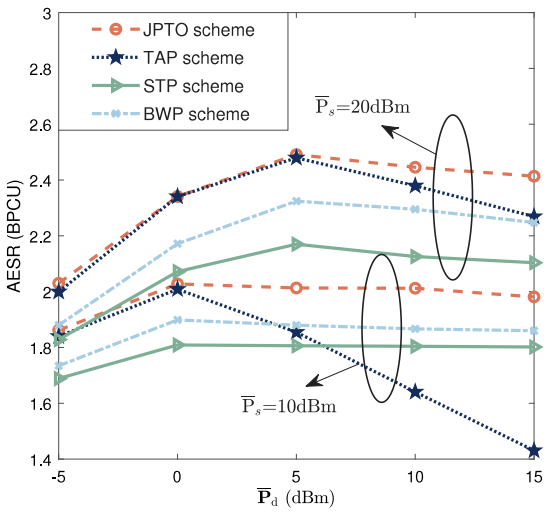


Fig. 8. AESR versus the average jammer power under different schemes.

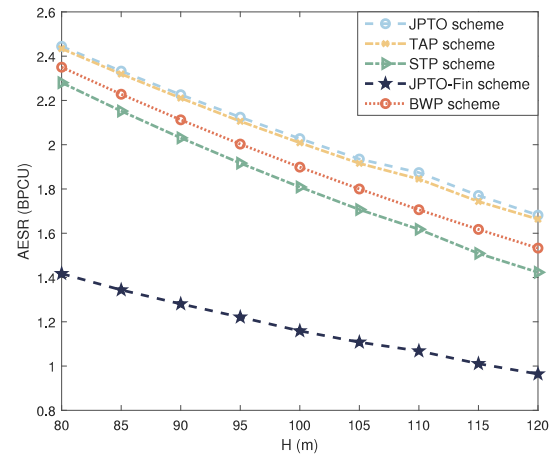


Fig. 9. AESR versus the UAVs' flight height under different schemes.

transmitting information later due to the finite packet length. This indicates that the nonnegativity of AESR is difficult to guarantee due to the latter two terms of the secrecy rate. Compared with the TAP scheme, it is observed that the proposed JPTO scheme and benchmark STP scheme and JTPO-Fin scheme can obtain the similar properties thanks to the power optimization, which have better communication performance. Specifically, the UAV S/J's power increases if the UAV S/J flies closer to the GR/Eve, and the power will decrease if the UAV S/J flies farther away from the GR/Eve and closer to the final position. Based on the above considerations, the transmit power reduction can save power and prevent intercepting when the UAV S/J flies straightly to the final position. In addition, for the TAP scheme that applies a constant power, the UAV S/J can approach GR/Eve by adjusting its trajectory, so as to enhance the communication performance and security performance of the mobile jammer-enabled secure UAV network, while it will bring about a longer arc trajectory.

Fig. 6 shows the AESR of different schemes versus flight period T. As expected, the proposed JPTO scheme outperforms other benchmark schemes and the JTPO-Fin scheme obtains the lowest AESR. This indicates that the significance of considering the joint optimization of the UAVs' trajectories and power under the optimal packet error rate. Moreover, it proves the importance of cooperative jamming UAV for improving the AESR. Specifically, the proposed JPTO scheme can achieve better AESR than the TAP scheme thanks to the power optimization. Similarly, compared with the STP scheme, the proposed scheme can obtain better performance due to its trajectory optimization. In addition, BWP scheme can achieve higher AESR performance than STP scheme due to the trajectory and power optimization of UAV J. This also can be indicated that the jamming UAV can effectively enhancing the secure performance. Note that the mobile jammer UAV can interfere with the Eve to achieve better security communication performance, this result suggests that the JPTO scheme can obtain better AESR than JTPO-Fin scheme by applying our proposed algorithm. Finally, it is observed that the AESRs of all the five schemes increase as T increases. The reason is that, a larger flying period will lead to a longer hovering time, resulting in a larger AESR and better secure communication performance.

Fig. 7 shows the AESR versus the packet error rate for different packet length and flight speed. It is observed that there exists a unique optimal packet error rate to maximize the AESR for given UAV S/J's transmit power and trajectory, which is consistent with the mathematical derivation and analysis in Section 3.5. In addition, the AESR increases with the packet length, this is because packet length has a significant impact on the system performance. Furthermore, the AESR increases rapidly with the flight period, this can be explained that a higher flight speed will lead to a longer hovering time, which can improve the communication performance and security performance accordingly.

Fig. 8 shows the AESR versus the average jamming power for different schemes. As $\bar{P}_d$ increases, the AESR of proposed JPTO scheme, TAP scheme, STP scheme and BWP scheme increase first and then decrease. This is because the excessive jamming power will interfere with GR, which will lead to the degrade of the system communication performance and security performance. Since there exists no jammer UAV, the benchmark JTPO-Fin scheme will not be considered here. It is found that the JPTO scheme can obtain better AESR as expected, which proves the superiority of the proposed optimization algorithm in this paper. Note that the AESR performance of BWP scheme can better than STP scheme, which can be demonstrated the significance effect of mobile jammer UAV. Moreover, the AESR of the TAP scheme drops sharply when $\bar{P}_d > 5$ dBm for $\bar{P}_s = 20$ dBm and $\bar{P}_d > 0$ dBm for $\bar{P}_s = 10$ dBm since each flight time slot utilizes equal jamming power. On this basis, when $\bar{P}_s = 10$ dBm, the AESR of STP scheme is better than the TAP scheme for $\bar{P}_d > 5$ dBm, while when $\bar{P}_s = 20$ dBm, the latter can obtain better security performance than the former. This indicates that, power optimization is of great significance to enhance the AESR in lower source power, while trajectories optimization is more effective in higher source power.

Fig. 9 shows the performance differences between the proposed JPTO scheme, TAP scheme, STP scheme, JPTO-Fin scheme and BWP scheme for different flight height. In this figure, AESR decreases monotonically when the H increase from 80 to 120 m. This is because that the secure communication quality between UAV and legitimate ground users becomes worse as the flight height increases. Moreover, due to the transmit power and trajectories optimization, our proposed scheme has better AESR performance than other four benchmark schemes, which can effectively address the AESR maximization problem. Noting that the JPTO-Fin scheme has the worst system performance among the four benchmark schemes, it can be verified the friendly mobile jammer has a considerable impact on improving the secure communication performance. By comparing the TAP scheme, STP scheme and BWP scheme, we further note that the transmit power optimization has a more significant influence on enhancing the secure communication performance than trajectories optimization. Therefore, the proposed alternating optimization algorithm can maximize the AESR of mobile jammer enabled secure UAV communication network.

5. Conclusions

This paper studies the mobile jammer enabled secure UAV communication within finite packet length. Subject to the peak/average

power constraints and UAVs' mobility constraints, the AESR performance is maximized by jointly optimizing the trajectories and transmit power of UAVs under the optimal packet error rate. Considering that the formulated problem is non-convex and complex to be solved, we firstly decompose the power and trajectories optimization problem into four sub-problems, i.e., the source UAV's power optimization, jammer UAV's power optimization, source UAV's trajectory optimization and jammer UAV's trajectory optimization, respectively. On this basis, we convert these four nonconvex subproblems into convex optimization problems by applying the first-order Taylor expansion. Then, the concavity of the packet error rate is analyzed comprehensively. Finally, an efficient alternating optimization algorithm for AESR problem based on Bisection method and SCA technology is proposed to address the five subproblems iteratively. Simulation results demonstrate that our proposed scheme can obtain evidently better security communication performance than the other four benchmark schemes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

[1] Gupta A, Trivedi A, Prasad A. Multi-UAV deployment for NOMA-enabled wireless networks based on IMOGWO algorithm. AEU – Int J Electron Commun 2022;153:154291.
[2] Cao D, Yang W, Li N. Throughput enhancement of buffer-aided multi-hop UAV relaying wireless networks. AEU – Int J Electron Commun 2021;131:153596.
[3] Li Y, Wang W, Gao H. Air-to-ground 3D channel modeling for UAV based on Gauss–Markov mobile model. AEU – Int J Electron Commun 2020;114:152995.
[4] Xiao Z, Shu J, Jiang H. Multi-objective parallel task offloading and content caching in D2D-aided MEC networks. IEEE Trans Mobile Comput 2022. http://dx.doi.org/10.1109/TMC.2022.3199876.
[5] Li X, Li J, Liu Y. Residual transceiver hardware impairments on cooperative NOMA networks. IEEE Tran Wireless Commun 2020;19(1):680–95.
[6] Zeng Y, Zhang R, Lim TJ. Wireless communications with unmanned aerial vehicles: Opportunities and challenges. IEEE Commun Mag 2016;54(5):36–42.
[7] Wang Z, Guo J, Chen Z, Yu L, Wang Y. Robust secure UAV relay-assisted cognitive communications with resource allocation and cooperative jamming. J Commun Netw-S Kor 2022;24(2):139–53.
[8] Wang Z, Zhou F, Wang Y, Wu Q. Joint 3D trajectory and resource optimization for a UAV relay-assisted cognitive radio network. China Commun 2021;18(6):184–200.
[9] Guo X, Li B, Zhang RZhaiD. Performance analysis and optimization of a UAV-enabled two-way relaying network under FSMH, NC, and PNC schemes. IEEE Internet Things J 2021;8(24):17802–16.
[10] Hu Z, Zeng F, Xiao Z. Computation efficiency maximization and QoE-provisioning in UAV-enabled MEC communication systems. IEEE Trans Netw Sci Eng 2021;8(2):1630–45.
[11] Zeng F, Hu Z, Xiao Z. Resource allocation and trajectory optimization for QoE provisioning in energy-efficient UAV-enabled wireless networks. IEEE Trans Veh Technol 2020;69(7):7634–47.
[12] Jiang H, Dai X, Xiao Z. Joint task offloading and resource allocation for energy-constrained mobile edge computing. IEEE Trans Mobile Comput 2022. http://dx.doi.org/10.1109/TMC.2022.3150432.
[13] Li X, Zhao M, Zeng M. Hardware impaired ambient backscatter NOMA systems: Reliability and security. IEEE Trans Commun 2021;69(4):2723–36.
[14] Li X, Zhao M, Liu Y. Secrecy analysis of ambient backscatter NOMA systems under I/Q imbalance. IEEE Trans Veh Technol 2020;69(10):12286–90.
[15] Gopala PK, Lai L, Gamal HE. On the secrecy capacity of fading channels. IEEE Trans Inform Theory 2008;54(10):4687–98.
[16] Zhou Y, Pan C, Yeoh PL, Wang KZ. Secure communications for UAV-enabled mobile edge computing systems. IEEE Trans Commun 2020;68(1):376–88.
[17] Zhou X, Yan S, Hu J, Sun J, Li J, Shu F. Joint optimization of a UAV's trajectory and transmit power for covert communications. IEEE Trans Signal Process 2019;67(16):4276–90.
[18] Sachs J, Wikstrom G, Dudda T, Baldemair R, Kittichokechai K. 5G radio network design for ultra-reliable low-latency communication. IEEE Netw 2018;32(2):24–31.
[19] Wu Q, Zeng Y, Zhang R. Joint trajectory and communication design for multi-UAV enabled wireless networks. IEEE Trans Wireless Commun 2018;17(3):2109–21.
[20] Zhang S, Zhang H, He Q, Bian K, Song L. Joint trajectory and power optimization for UAV relay networks. IEEE Commun Lett 2018;22(1):161–4.
[21] Wu Q, Zeng Y, Zhang R. Joint trajectory and communication design for UAV-enabled multiple access. In: 2017 IEEE global communication conference. 2017, p. 1–6.
[22] Wang W, Tang J, Zhao N. Joint precoding optimization for secure SWIPT in UAV-aided NOMA networks. IEEE Trans Commun 2020;68(8):5028–40.
[23] Dong R, Wang B, Cao K. Security enhancement of UAV swarm enabled relaying systems with joint beamforming and resource allocation. China Commun 2021;18(9):71–87.
[24] Pang X, Sheng M, Zhao N. When UAV meets IRS: Expanding air-ground networks via passive reflection. IEEE Wirel Commun 2021;28(5):164–70.
[25] Zhang G, Wu Q, Cui M, Zhang R. Securing UAV communications via joint trajectory and power control. IEEE Trans Wirel Commun 2019;18(2):1376–89.
[26] Zhou X, Wu Q, Yan S, Shu F, Li J. UAV-enabled secure communications: Joint trajectory and transmit power optimization. IEEE Trans Veh Technol 2019;68(4):4069–73.
[27] Pan C, Ren H, Deng Y, Elkashlan M, Nallanathan A. Joint blocklength and location optimization for URLLC-enabled UAV relay systems. IEEE Commun Lett 2019;23(3):498–501.
[28] Wang Y, Zhou X, Zhuang Z, Sun L, Qian Y, Lu J. UAV-enabled secure communication with finite blocklength. IEEE Trans Veh Technol 2020;69(12):16309–13.
[29] Wang H, Yang Q, Ding Z, Poor HV. Secure short-packet communications for mission-critical IoT applications. IEEE Trans Wirel Commun 2019;18(5):2565–78.
[30] Grant M, Boyd S. CVX: Matlab software for disciplined convex programming, version 2.1. Mar. 2014.
[31] Hu H, Zhang H, Liang C. On the spectrum- and energy-efficiency tradeoff in cognitive radio networks. IEEE Trans Commun 2016;64(2):490–501.
[32] Yeung RW. A first course in information theory. Springer; 2002, p. 215–31.
[33] Li A, Zhang W. Mobile jammer-aided secure UAV communications via trajectory design and power control. China Commun 2018;15(8):141–51.